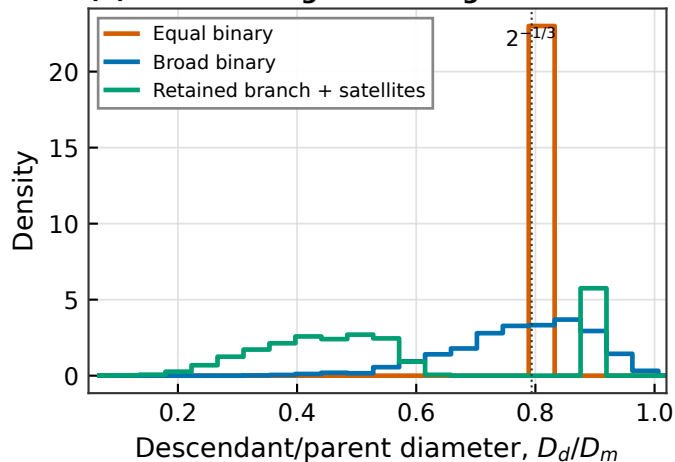
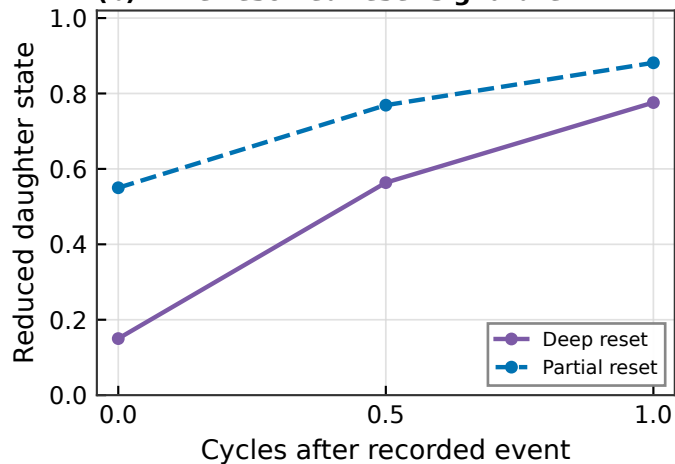
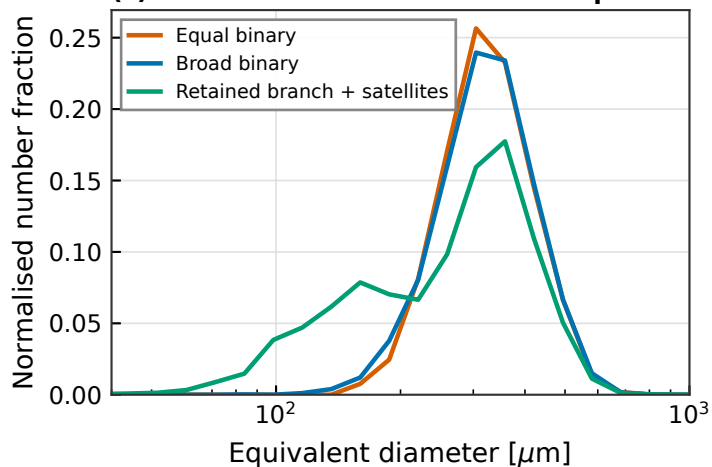


**(a) Paired lineage kernel signature****(b) Time-resolved reset signature****(c) Distribution after known event pulses****(d) Prospective protocol adequacy**